

The Approach of ACARS Data Encryption and Authentication

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Abstract—With the increasing number of aircraft, the airlines and civil aviation organization are relying on ACARS (Aircraft Communications Addressing and Report System) for air traffic and control. Unfortunately, automated information exchange between aircraft and ground entity also increase the system vulnerability and may severely impacts the safety of a flight. This paper introduce a End-to-End architecture which employ message encryption and authentication to protect the security between aircraft and airlines and thus permit ACARS datalink messages to be exchanged in a secure manner.

KeyWords—ACARS, security, message authentication, data encryption

I. INTRODUCTION

With the development of civil aviation, the airlines need efficient data exchange between flight and ground service center. The International Civil Aviation Organization (ICAO) proposes the conception of “free flight” which depends on the transition of Air Traffic Control (ATC) voice message traffic to data link message [1]. ACARS is a data link which exchanges character-oriented data between aircraft system and ground system. It works in an automated manner, which can reduce crew workload and improve data trustiness.

The ACARS message is transferred over open RF channels in human readable forms [2], low cost equipment or software enable anyone to monitor and process ACARS message (such as acarsd). It is possible to determine aircraft type, position, and projected track, etc. The airline would like to protect this information to maintain competitive advantage. In addition, the military is beginning to exploit ACARS for their aircrafts [1] [3]; the potential for unauthorized interception of ACARS message for military also increases. This paper will show a secure ACARS with End-to-End architecture and adopt Public Key Infrastructure (PKI) techniques.

II. SECURE ACARS FRAMEWORK

The secure ACARS framework specifies security architecture and scheme for ACARS data link, ACARS can be simply described as fig. 1[4]

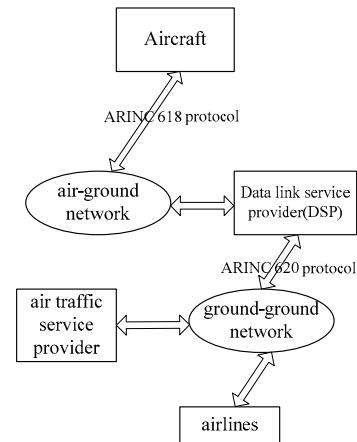


figure1. ACARS components

In fig. 1, the air-ground network maybe VHF, HF or satcom based on the position of aircraft and the useable service. ARINC and SITA are the two dominant service providers who carry the bulks of ACARS data through their private networks [3] (the ground-ground network in fig. 1). All uplink and downlink messages are sent via a central processing system (CPS), maintained by the DSP, which performs message formatting, routing, and relaying functions using the information contained in the ACARS message header [3] [4] as shown in TABLE1. (Take downlink message for example, the uplink is similar) Ground hosts are typically interfaced with the DSP via private network.

TABLE I. ACARS Downlink Message Format [3] [4]

Filed Name	SOH	MODE	A/C Registration	ACK/ NAK	Label	DBI	STX	Message Sequence Number	Agency ID	Flight NO.	User Data text
Size (byte)	1	1	7	1	2	1	1	4	2	4	0-210
example		2	.N123XX		Q5	A		M01A	AA	1234	Text
Message header used by DSP for message routing(unprotected)								User data that be protected			

A. Security architecture

In general, there are two security architecture to implement [2], DSP-based architecture and End-to-End architecture, which both provide a security envelope that provides data encryption and message Authentication service. The DSP-based only protects ACARS message as they traverse air-ground sub-network, for the End-to-End security architecture, the message can be protected both in the air-ground and ground-ground sub-network, so this is particularly suited for protection ACARS message that contain airline-proprietary or sensitive information, such as information linked to individuals. This paper will adopt End-to-End architecture.

B. Security scheme

To construct a secure ACARS, we have to assure the clear text can not be read and changed when they are transmitted, moreover, the communication peers must know whether compeer is who it claims, and so we should concern [3]:

- Privacy: make sure that the message is protected from disclosure during transmission.
- Integrity: provide communication entity with assurance that the message is protected against modification (such as insert, delete, or substitution of message).
- Authentication: provide message recipient with assurance that the sender of the message is as-claimed.

The next section will detail how to ensure the three items described above.

III. PROTECTION MODE

The protection mode specifies the secure configuration of ACARS, and how to produce key for data encryption and message authentication.

A. Key derive

For the protection of privacy and integrity of message, encryption and digital signature or message authentication can be taken into account. For data encryption, both symmetric and asymmetric cryptography can be employed to protect ACARS message, but they have their own advantages as well as disadvantages respective [3]: In the symmetric cryptography scheme, both the sender and recipient use the same key encrypt and decrypt the data, as such, they requires that every pair of communication parties to have a unique secret key that is protected from unauthorized access, this is difficult to achieve for a large number of users. Although asymmetric scheme overcome the key distribution problem, it requires a large key size to provide the same level of security, therefore, in bandwidth-limited ACARS it is also inappropriate.

ATN Security SARPs specified a hybrid scheme for aeronautical communication to take advantage both symmetric and asymmetric scheme[5]: Before the communicate data exchanges, each entity uses elliptic curve

Diffie-Hellman (ECDH) to derive a shared secret parameter with operating as follow [1] [2]:

- The entity selects an elliptic curve E and a point P on E and their private key d (a random integer).
- Compute their public key d_aP d_gP respective, where d_aP mean the public key of aircraft and d_gP mean public key of ground.
- Two entities exchange their public keys, and then derive a shared secret parameter use ECDH: $Z = d_a(d_gP) = Z' = d_g(d_aP)$

The ECDH scheme can make sure the equation of Z with Z' and prevent an unauthorized entity from deriving Z even though the exchange of public key is intercepted (the private key can not be got).

In a secure system, the shared parameter Z typically does not have the desired properties to be used directly as cryptographic keys, it is generally noted as “raw key materials”[7], so a key derivation function (KDF) is needed to get the message Authentication key and data encryption key:

$$Key_{MAC} = SHA-256(Z || R || 0x01) \quad (1)$$

$$Key_{enc} = SHA-256(Z || R || 0x02) \quad (2)$$

Where SHA defines the Secure Hash Algorithm and “||” denotes concatenation, R is a random value and the 16-bit integer “01” and “02” are used to ensure the SHA-256 operation will yield different values for the Key_{MAC} and Key_{enc} each time. The result of SHA-256 is a random binary sequence that is 256 bit; Key_{enc} is used as the key of Advanced Encryption Standard (AES) for providing encryption/decryption of ACARS application data. As detailed above, the digital signature with asymmetric key is not the best choice so the Key_{MAC} is used to produce Message Authentication Code which append to the message for integrity check. The MAC can be produced by:

$$MAC = SHA-256(encrypted\ data || Key_{MAC}) \quad (3)$$

So a wholly protected ACARS message can be described as table 2:

TABLE II. Protected message		
Message header (unprotected)	Encrypted data	MAC

B. Secure application data transfer

Once Key_{MAC} and Key_{enc} have been derived, aircraft and ground entities can use them to exchange ACARS messages in a secure way as fig. 2[2].

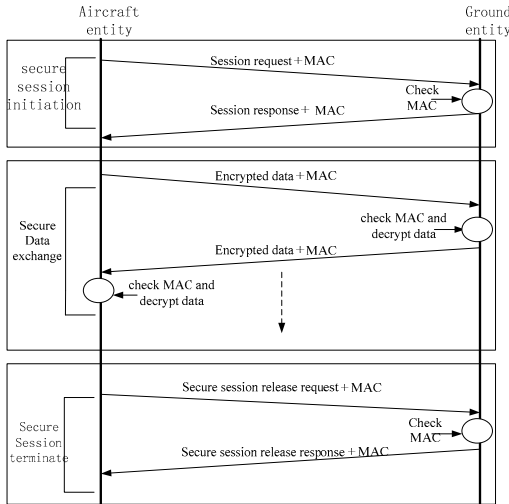


figure2. secure data exchange

When one entity receives a message from

TABLE III. Payload encoding table [2]

character			character			character			character		
ASCII	HEX	6 bit code	ASCII	HEX	6 bit code	ASCII	HEX	6 bit code	ASCII	HEX	6 bit code
sp	0x20	000000	0	0x30	010000	@	0x40	100000	P	0x50	110000
!	0x21	000001	1	0x31	010001	A	0x41	100001	Q	0x51	110001
"	0x22	000010	2	0x32	010010	B	0x42	100010	R	0x52	110010
#	0x23	000011	3	0x33	010011	C	0x43	100011	S	0x53	110011
\$	0x24	000100	4	0x34	010100	D	0x44	100100	T	0x54	110100
%	0x25	000101	5	0x35	010101	E	0x45	100101	U	0x55	110101
&	0x26	000110	6	0x36	010110	F	0x46	100110	V	0x56	110110
'	0x27	000111	7	0x37	010111	G	0x47	100111	W	0x57	110111
(	0x28	001000	8	0x38	011000	H	0x48	101000	X	0x58	111000
)	0x29	001001	9	0x39	011001	I	0x49	101001	Y	0x59	111001
*	0x2A	001010	:	0x3A	011010	J	0x4A	101010	Z	0x5A	111010
+	0x2B	001011	;	0x3B	011011	K	0x4B	101011	LF	0x0A	111011
,	0x2C	001100	<	0x3C	011100	L	0x4C	101100	\	0x5C	111100
-	0x2D	001101	=	0x3D	011101	M	0x4D	101101	CR	0x0D	111101
.	0x2E	001110	>	0x3E	011110	N	0x4E	101110	^	0x5E	111110
/	0x2F	001111	?	0x3F	011111	O	0x4F	101111		0x7C	111111

Based on table 3, each ACARS character can be transferred to a 6-bit stream and then use the bit stream for data encryption process, the recipients can also decode the bit stream to character based on this table.

- Message encoding/decoding: after encryption scheme, the produced bit stream may not be transmit directly in ACARS, they have to be encoded as ACARS-readable character, this process uses a message encode table which is almost the same with payload encoding table, except substitute "LF" and "CR" with "[" and "]" respective[2]. A sending entity performs the following message encoding process:

communication peer, they verifies the data integrity by compute MAC with the KeyMAC and compares with that appendix on the message, if the message is authenticated, it then decrypts the encrypted message, otherwise it will drop the message and take no account of it, thus protecting the data privacy and integrity.

C. Other detail

As we know the ACARS messages are character-oriented (uses ISO-5 character table) and the message length has a limit. In addition, the encryption process is bit-oriented, also it will produce bit stream. After encryption, the new character derived from bit stream may not be transferred in ACARS (not included in ISO-5) and the new message length may go beyond the max length, so message encoding and compression need to implement.

- Payload encoding/decoding: this process can transfer character to bit stream base on table 3, and encode the 8-bit to 6-bit thus reducing the data length by 25% [2].

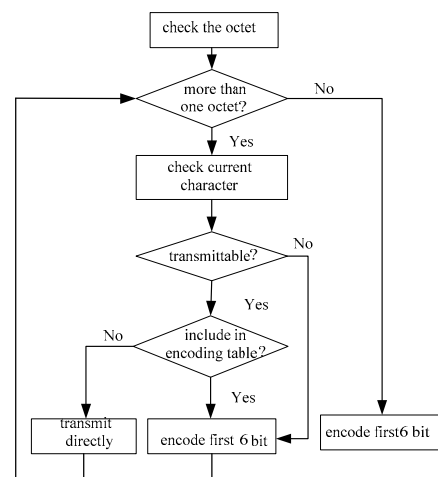


figure3. message encoding

Note: if an octet is transmittable, we have to verify whether this character is included in encoding table, otherwise it will confuse the recipient that a received character is transmitted directly or is coded from a 6-bit data. Now we can depict the process of communication with fig. 5.

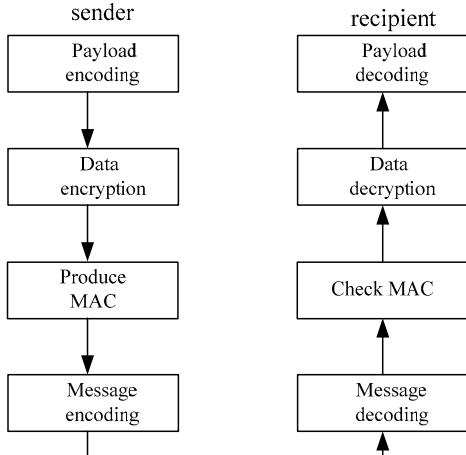


figure4. communicate process

IV. TEST RESULTS

Test system framework is shown as fig. 6. In the End-to-End architecture, the communication peers are aircraft and airline. Air-ground subnet is simulated with RS232 communication and ground-to-ground sub network is TCP/IP net. Eavesdrop can read the ACARS message as airlines; also it can serve as a pseudo aircraft and send ACARS message to airlines.

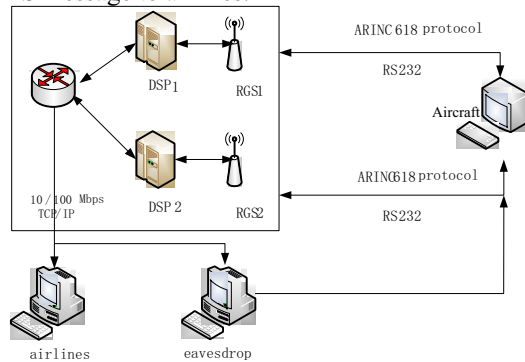


figure5. System framework

There will examine the Encryption and Authentication mechanism respective. First, two plain ACARS messages are:

Message1: M04AAF0185VHHHLFPG15541006
 Message2: M31AMU5201POSCAS 248,LAT N22.033,LON E113.307,ALT4644,UTC093624

After encrypted, they will be:

Message1: A]P+f+w-+|MaF 7#1]P\(+\$pP
 Message2:A^ZP'*GT4PIYNJK5UPb0Dq6D7A'L,+HV
 O|Yn+T1{Do(!6WnF#-Sdw@'^-G4(S0

The message can not be read even they are received by eavesdrop. AES with 256 bit key can protect the security of message and stand cryptanalysis. Under this condition the privacy can be protected. For the message integrity, airlines will check the MAC and determine whether a message is modified.

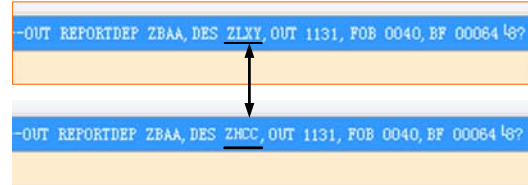


figure6. message modify

Fig. 7 shows that an ACARS out-report message has been substitute characters "ZHCC" for "ZLXY", this field means the destination of the flight. In a natural condition, the airlines should receive a message and extract every field as shown in fig. 8.



figure7. message in natural condition

When the message authenticate mechanism is not on work, the airlines will regard the pretend message as valid and treat the destination as real, shown in fig. 9.

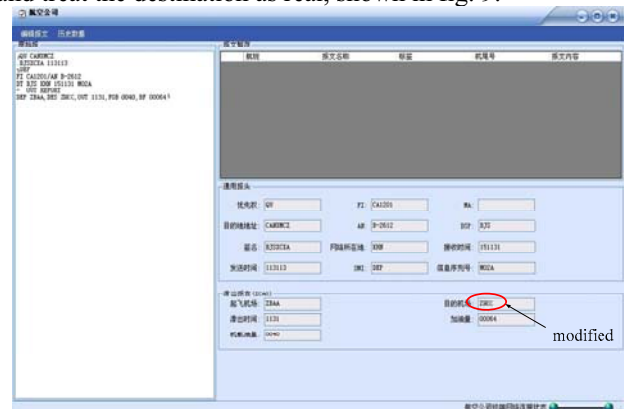


figure8. received message without authentication

If the message authentication mechanism is used to check every message that has been received, this modified message will be dropped. If the eavesdrop sends message to

aircraft, because it can not get the airline’s MAC so that the content it send will be check out by the aircraft. As shown in fig. 10 in the red field.

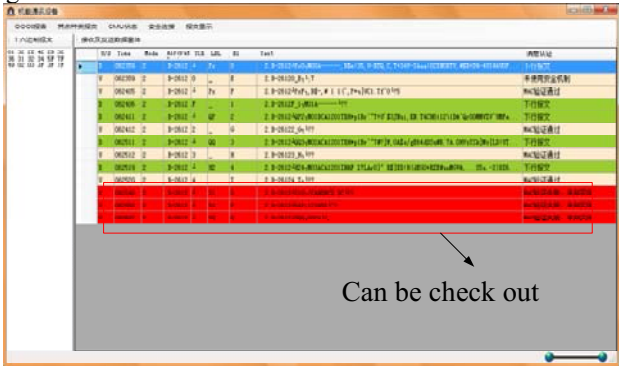


figure9. MAC checked by aircraft

In order to implement this scheme, we must designate an entity as the Certificate Authority (CA), to produce and distribute the asymmetric key pairs and keep a Certificate Revocation Lists (CRL), which should be used by the communicating peers real-time, but in our system the key pairs it pre-load in the aircraft and ground entities. Furthermore, to build a certificate such as X.509 is necessary for CA. To get better time and less memory costs, one or multiple data compress algorithms that subsequent to data encryption and prior to the message encoding process should be introduced to the system

With the data encryption and message authentication, the system can protect message privacy and data integrity for ACARS, furthermore, End-to-End architecture also protects the security of whole link. As the cryptographic algorithms and authenticate specified for ACARS is derived from ATN standards, it is easy to migrate to ATN and protect the investment.

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